CREDIT RISK UNDERWRITING ARCHITECTURE

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Problem & Solution: High default misclassifications jeopardize portfolio yield; this pipeline deploys an interpretable Glass-Box EBM to capture ≈70% of defaults while maintaining full regulatory auditability.

01 • DISCOVERY

Business Problem

86% Defaults missed by naive 0.50 cutoff, driving severe uncollectible write-offs.

- **Severe Imbalance:** 78:22 split heavily biases training.
- **Naive Cutoff Trap:** 0.50 threshold misses >85% write-offs.
- **Target Mandate:** Capture ≥70% default risk reliably.

02 • METHODOLOGY

1. Raw Ingestion — INPUT
1.3M+ LendingClub loan records

2. Signal Distillation — TRANSFORM
23 features (Log1p, Yeo-Johnson transforms)

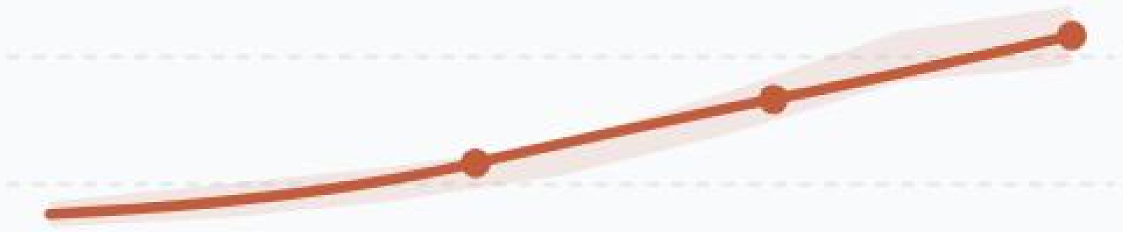
3. Validation Engine — VALIDATION
Stratified 10-fold CV with leak-free holdouts

4. Dual-Track Modeling — EVALUATION
GBDT Ensembles vs. Glass-Box EBM

03 • FINAL DECISION — CHAMPION

Glass-Box EBM

Additive Shape Curve f(DTI) — EXACT MATH (GA^2M)



- **100% Auditable:** Exact additive splines with zero opacity.
- **FCRA / ECOA:** Direct adverse-action reason codes.

0.7265	**69.1%**	**40,140+**	**78.2%**
TEST ROC-AUC	DEFAULT RECALL	BAD LOANS BLOCKED	GOOD LOAN RETENTION
Glass-Box Champion Performance	Captured at Strategy Cutoff (t=0.22)	Out of 58,120 Test Defaults Prevented	Preserved Volume Under Strategy